

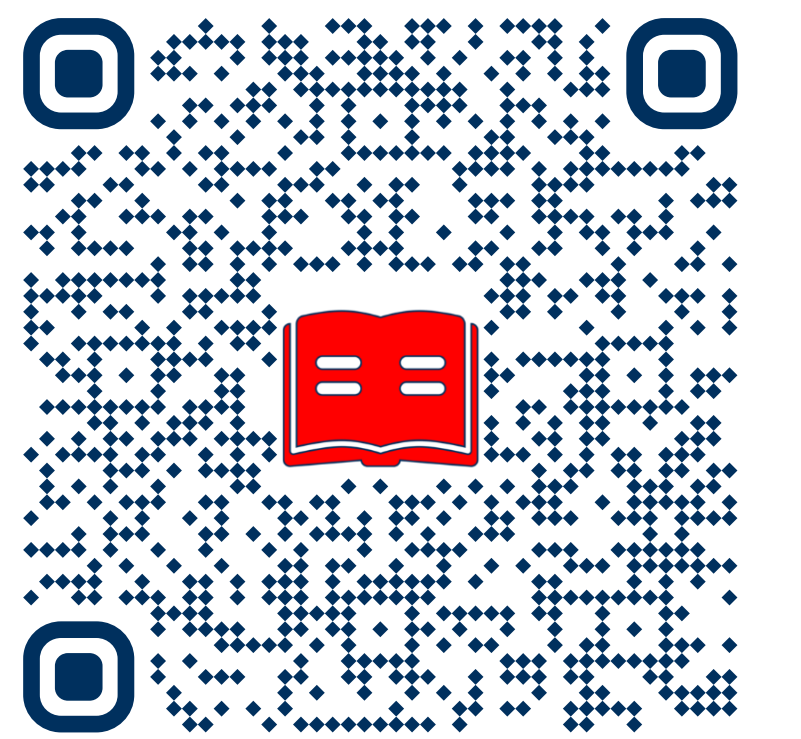


ENHANCED INTERFERENCE BETWEEN PAVLOVIAN AND INSTRUMENTAL CONTROL IN ALCOHOL USE DISORDER BUT NOT SMOKING

Louis Thill

Hao Chen, Christian Baeuchl, Andreas Heinz & Michael N. Smolka

Department of Psychiatry and Psychotherapy, Technische Universität Dresden, Dresden, Germany



Material & Contact Info

Pavlovian-to-instrumental transfer (PIT)

Pavlovian conditioned stimuli (CS) can influence the intensity & accuracy of instrumental behavior¹

Pos. CS → Activation + Approach
Neg. CS → Inhibition + Avoidance



Enhanced PIT in individuals with risky drinking^{2,3}
+
Associated with consumption increases²

Enhanced PIT in alcohol-dependent patients^{4,5}
+
Associated with relapse after detoxification⁵

Research Question 1:

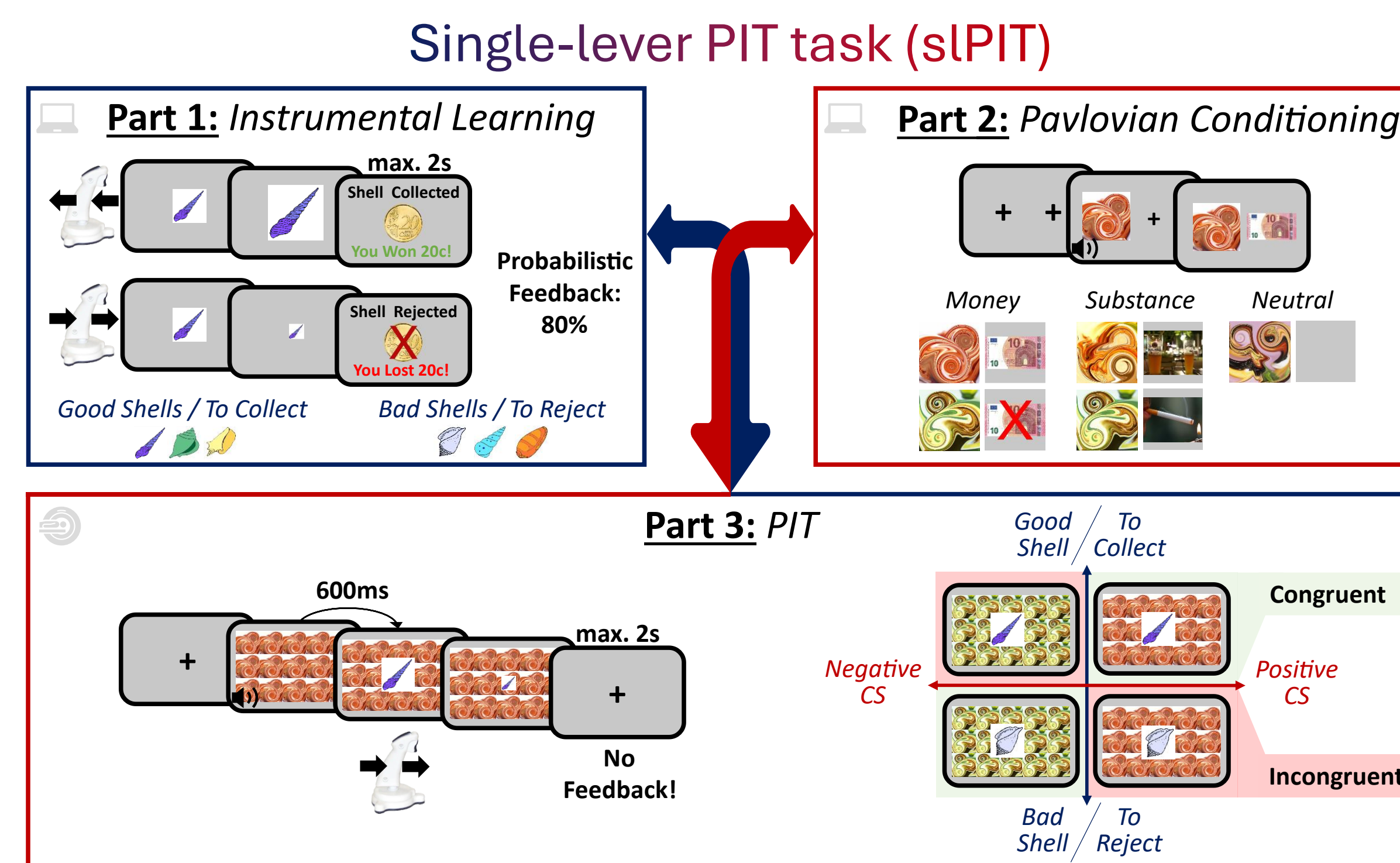
Can enhanced PIT effects be replicated in a broader non-treatment-seeking AUD sample?

Research Question 2:

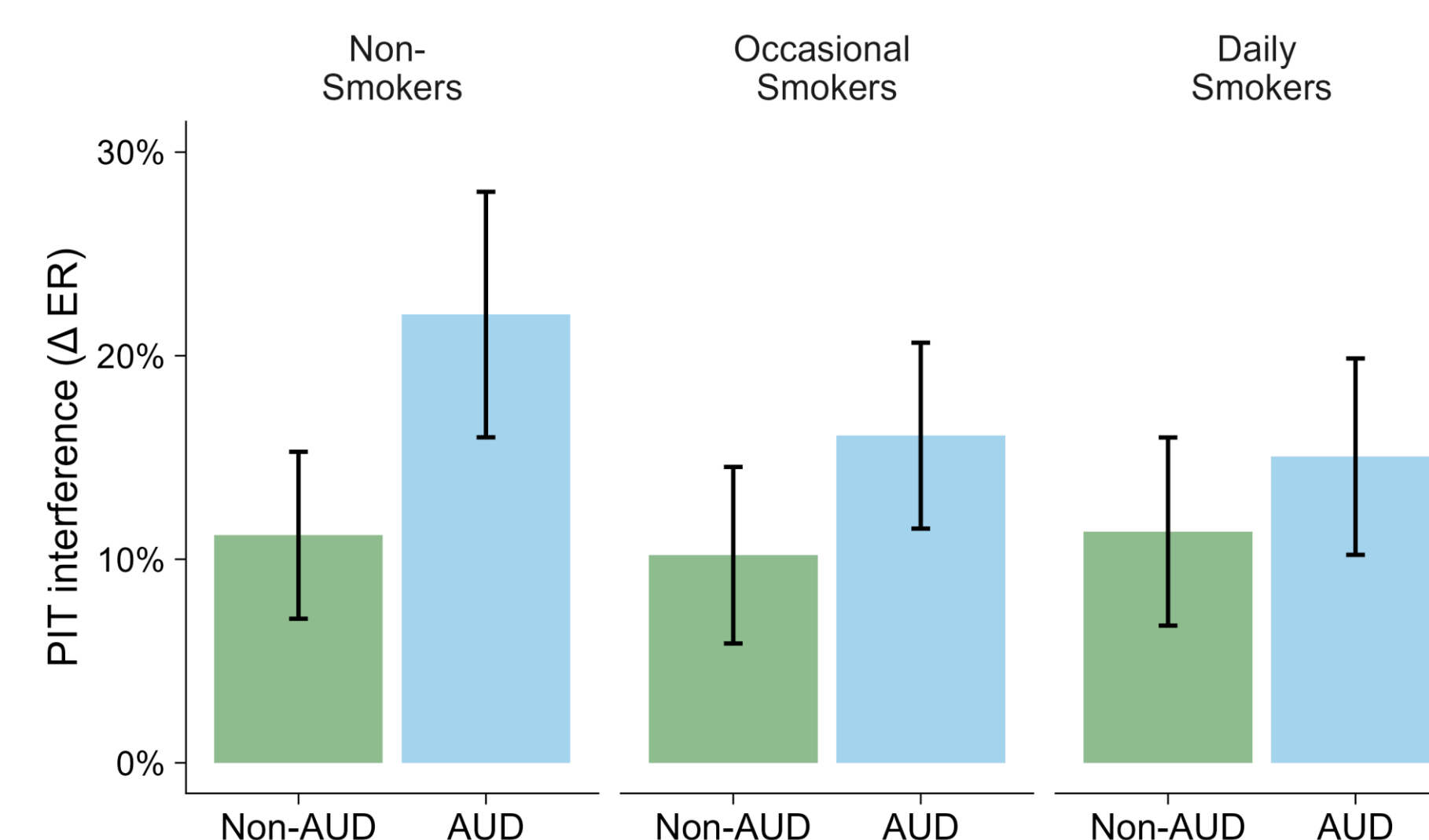
Do PIT effects increase with increasing smoking status? (*transdiagnostic hypothesis*)

Research Question 3:

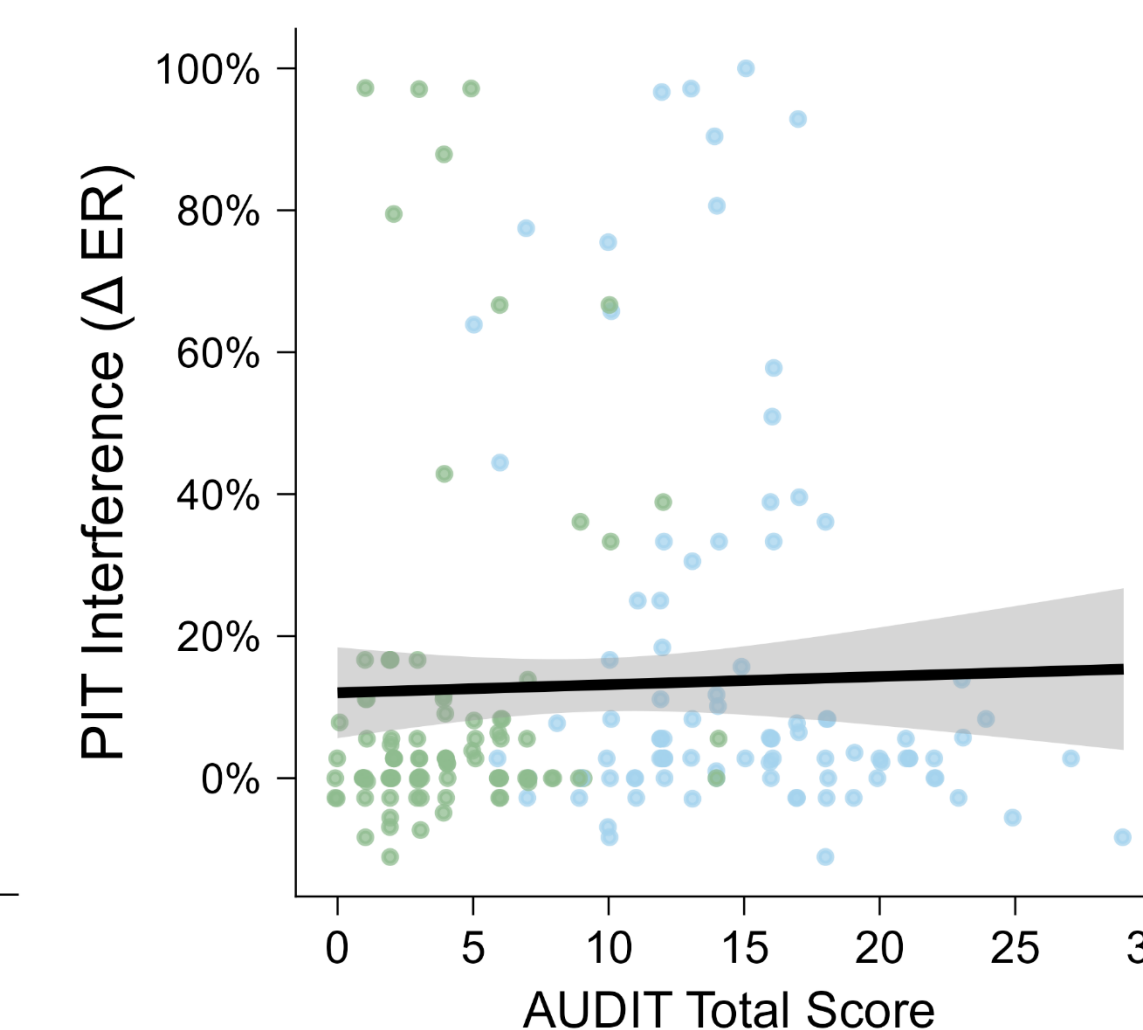
Is there a synergistic interaction between AUD and smoking with respect to PIT?



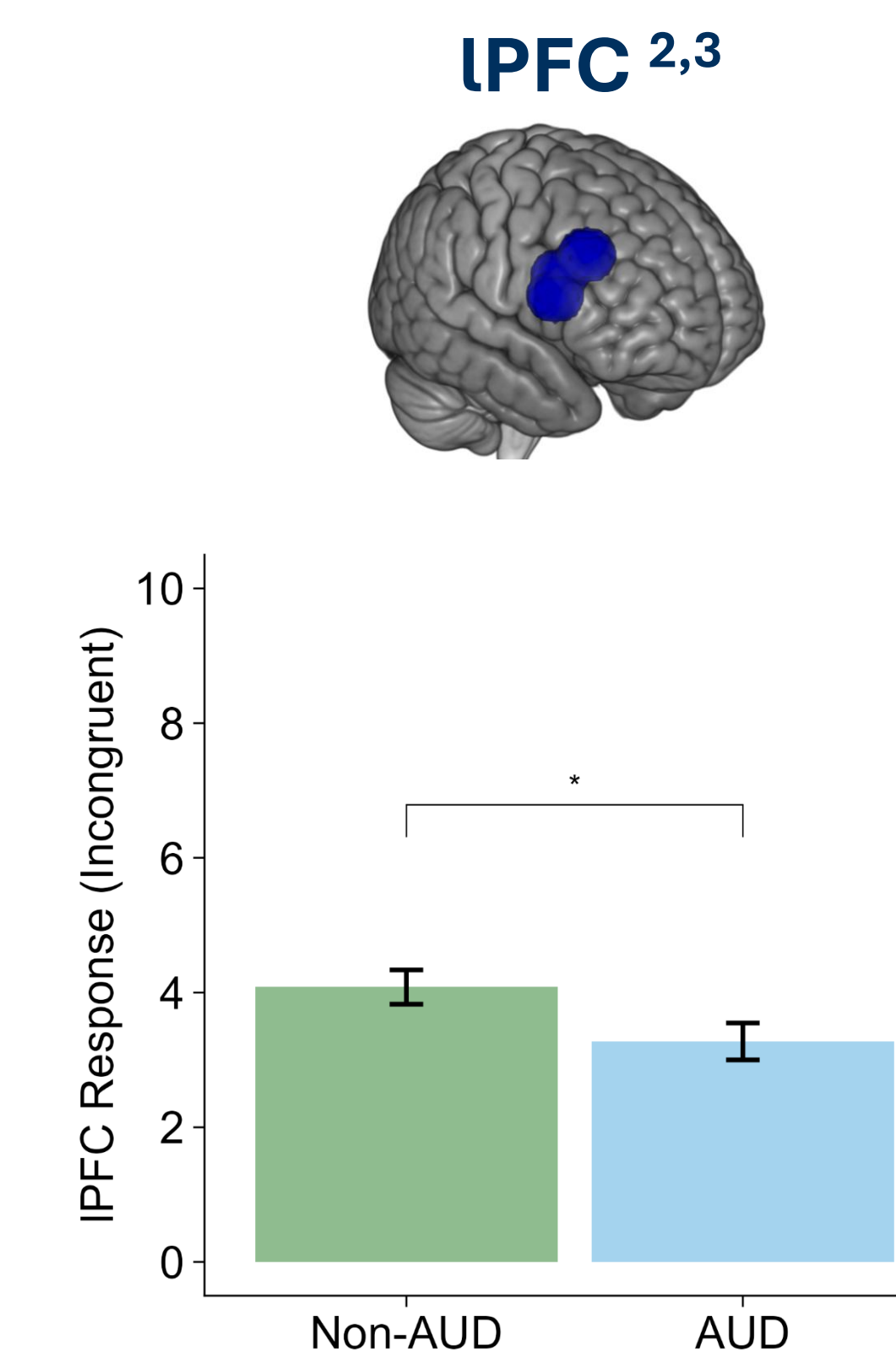
PIT-Interference:
 $\Delta ER = ER_{\text{incongruent}} - ER_{\text{congruent}}$



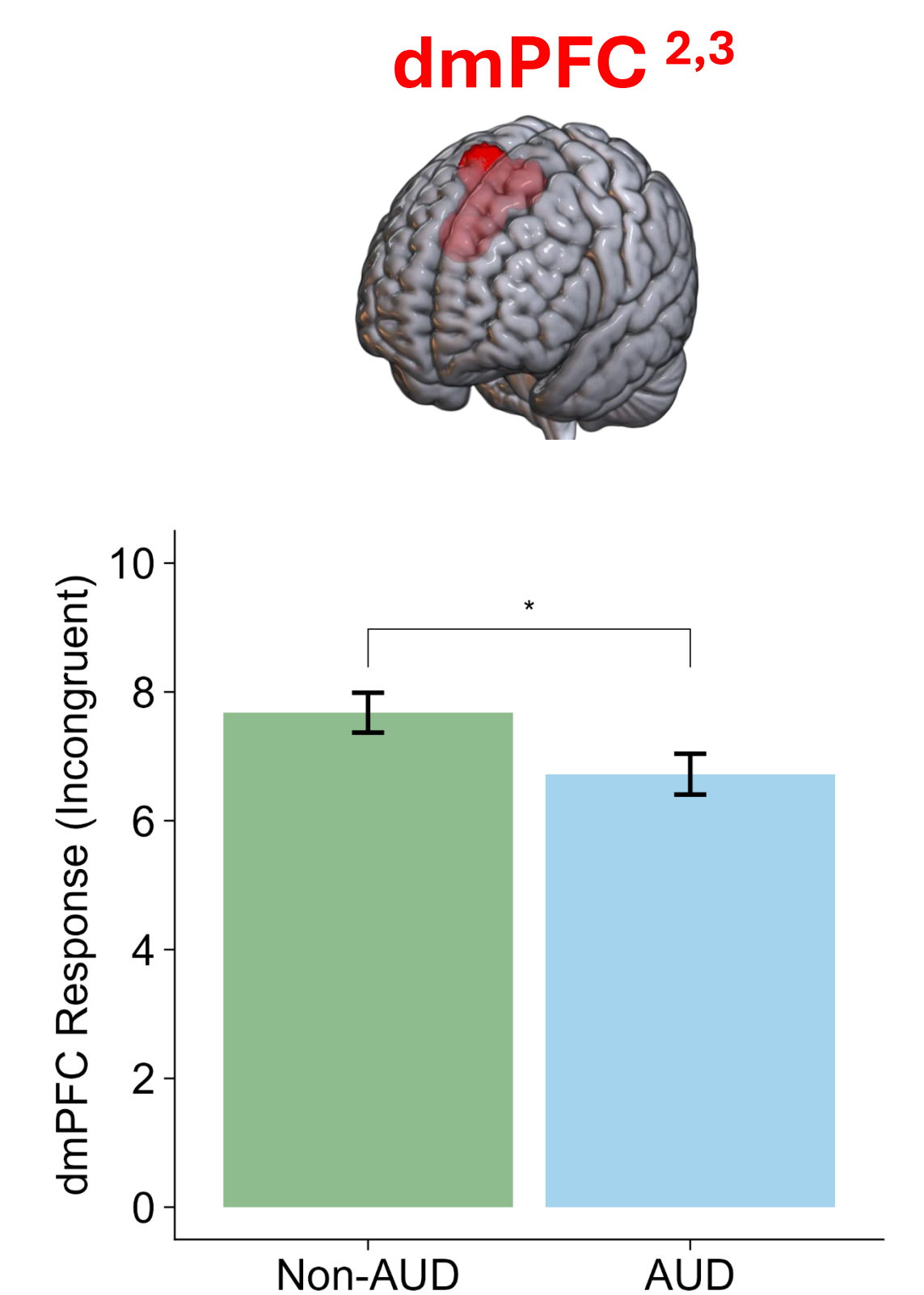
AUD: $\eta_p^2 = 0.02$; $p = .03$
Smoking: $\eta_p^2 = 0.001$; $p = .83$
AUD x Smoking: $\eta_p^2 = 0.01$; $p = .37$



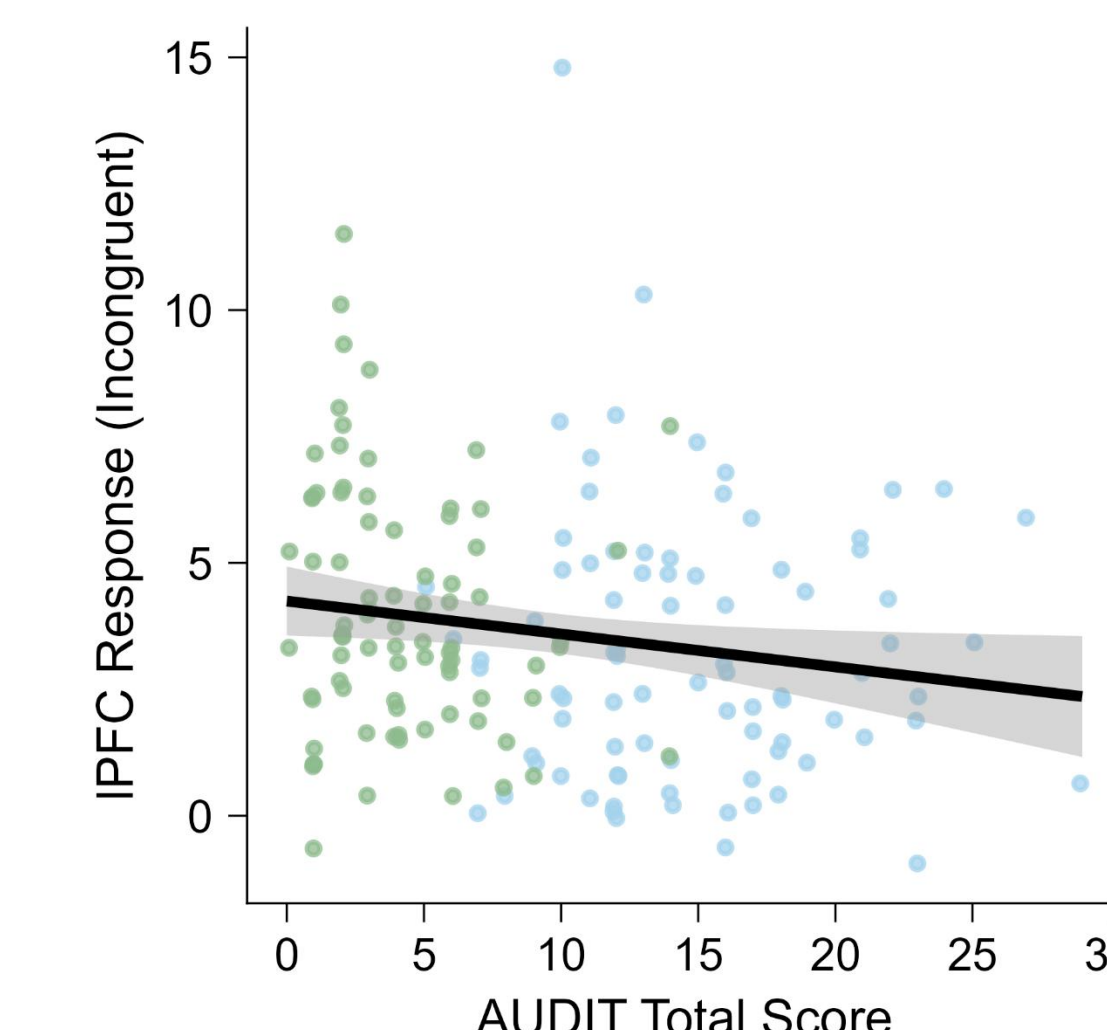
$\tau = .10$; $p = .03$



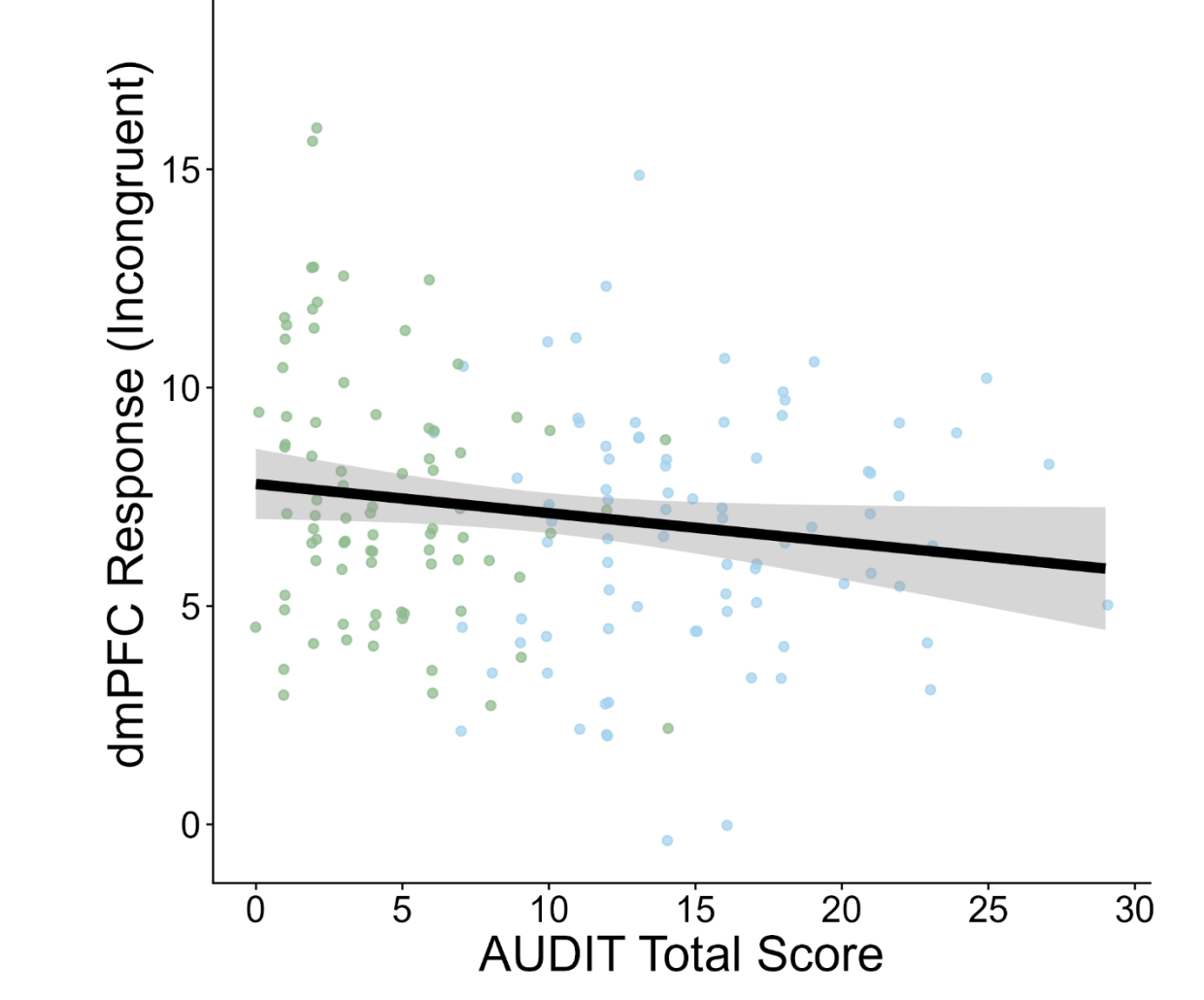
$d = .32$; $p = .02$



$d = .32$; $p = .02$



$r = -.17$; $p = .01$



$r = -.15$; $p = .03$

Pre-Screening Assessment		Basic Assessment	MRI Session 1	MRI Session 2	3 FUs & EMA	
		2x3 Stratification	slPIT	ftPIT	1 year	
N = 200			N	Age(y)	♀(%)	AUD criteria
Factor 1	AUD:		100	33.2	40.0	5.2
	Non-AUD:		100	29.8	42.0	1.1
Factor 2	Non-Smoker:		70	32.7	42.9	3.0
	Occasional Smoker:		72	29.8	40.3	3.2
	Daily Smoker:		58	32.1	39.7	3.1
						TUD criteria

Conclusion

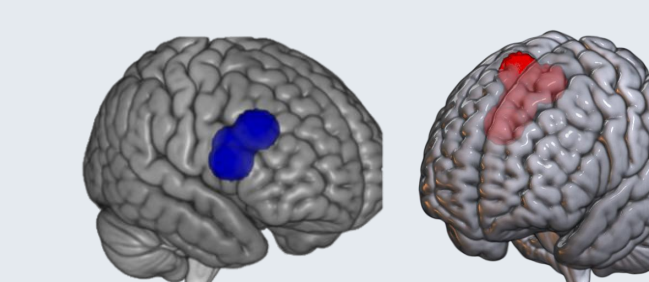
1

Enhanced PIT in non-treatment-seeking individuals with predominantly mild-to-moderate AUD (majority of AUD cases⁶)



2

Potentially related to reduced recruitment of prefrontal control regions during PIT conflicts



3

No evidence for increasing PIT with increasing smoking status or AUD x smoking interaction

